DATA ANALYTICS PROJECT

TVS Motor Manufacturing Quality, Defect & Production Analytics

End-to-end workflow using Excel • CSV • MySQL • Power BI

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GitHub: github.com/prithivikanth/prithivikanth-TVS-Motor-Manufacturing-Quality-Analytics

Portfolio demonstration • Synthetic practice dataset

Project Context & Data Disclaimer

What this project demonstrates

A structured workflow for preparing, validating, analysing and visualising manufacturing-style data.

What it does not claim

The datasets are synthetic practice data—not official TVS Motor Company records or verified factory results.

Project workflow

01

Excel workbook

02

Clean CSVs

03

MySQL

04

SQL validation

05

Power BI

Business Problem & Objectives

Business problem

Manufacturing data can be spread across production logs, defect logs and reference tables. Without consistent data modelling and KPI definitions, comparisons across plants, machines, products and shifts can be misleading.

Project objectives

Organise plant, machine and product reference data.

Analyse production quantities and recorded defects.

Explore defect categories, root-cause labels, downtime and rework.

Validate imports, row counts and quality flags.

Build a Power BI view and document the workflow.

Technology Stack

Microsoft Excel

Inspect the original workbook.

CSV files

Portable tabular datasets for the workflow.

MySQL 8 / Workbench

Relational tables, validation and SQL analysis.

Power BI Desktop

Data model, measures and interactive visuals.

GitHub

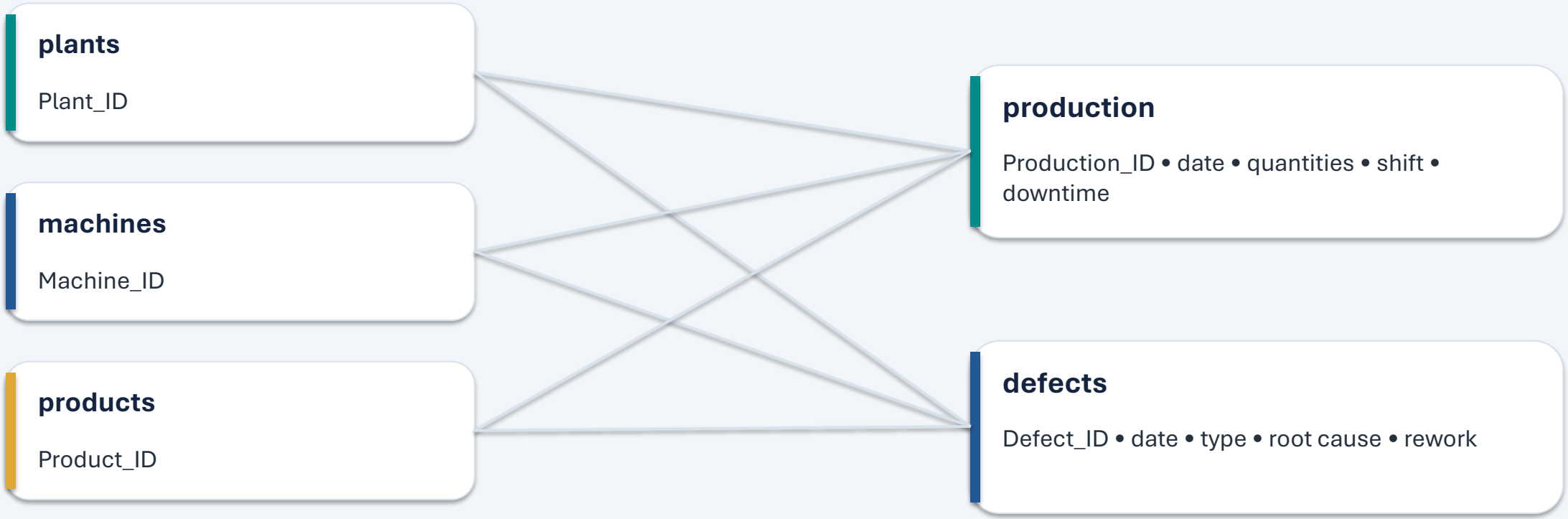
Versioned project files and documentation.

Dataset Inventory: Five Cleaned CSVs

CSV file	Role	Main fields / purpose
plants.csv	Dimension	Plant IDs, names and state
machines.csv	Dimension	Machine IDs, names and types
products.csv	Dimension	Product IDs, names and categories
production.csv	Fact table	Production quantity, defect quantity, shift, downtime, quality flag
defects.csv	Fact table	Defect type, root-cause label, quantity, rework, quality flag

The raw Excel workbook is kept separately. See the repository's Data Dictionary for every documented column, key and relationship.

Data Model & Relationship Logic



Caution: production and defects are separate fact tables. Direct row-level joins can multiply records and inflate totals; aggregate to a compatible grain before combining.

End-to-End Workflow

01
Inspect raw workbook

02
Review cleaned CSVs

03
Create MySQL schema

04
Load data

05
Validate counts & flags

06
Run SQL analysis

07
Refresh Power BI

Execution order matters: create → insert → verify → analyse. Check errors before proceeding.

SQL Scripts & Validation

Script	Purpose
01_create_tables.sql	Creates database and tables
02_insert_data.sql	Loads the practice records
03_verify_import.sql	Checks row counts, flags, dates and KPI targets
04_analysis_queries.sql	Runs analysis queries

Important safety warning

The table-creation script contains DROP DATABASE IF EXISTS tvs_manufacturing. It deletes and recreates that database. Do not run it against a database containing important work.

Validate actual SQL output before quoting any KPI in the presentation.

Validation Targets Documented in SQL

3,626,137

Production units

OK production rows

138,990

Production-table defect units

OK production rows

3.83%

Production-table defect rate

Defined as defect qty ÷ production qty

34.80 min

Average downtime

OK production rows

9,285

Defect-log units

OK defect rows

These figures are expected validation targets documented in the project SQL—not independent company facts. Re-run the validation queries and reconcile the Power BI measures before presenting them as results.

Keep production.Defect_Qty distinct from defects.Defect_Qty; the two fields come from separate fact tables.

Business Questions the Analysis Can Explore

How does production quantity vary by plant, machine, product and shift?

Which defect categories and recorded root-cause labels appear most often?

How is downtime distributed across plants or machines?

What share of defect records is marked as requiring rework?

How do results change when non-OK quality records are excluded?

Do Power BI measures reconcile with SQL validation output?

Report only patterns confirmed by actual query outputs and dashboard filters.

Power BI Dashboard Walkthrough

Page 1 — Overview

Show the main KPI cards and explain each metric’s numerator, denominator and quality filter.

Page 2 — Quality

Explain defect categories, defect rate and any quality comparisons that exist in the actual report.

Page 3 — Machines

Show machine-level comparisons and explain the filter context used.

Use screenshots stored in GitHub as backup. Replace these guide cards with real dashboard screenshots if you want a more visual deck.

Data Quality Checks & Limitations

Validation checklist

- Check row counts and duplicate IDs
- Validate foreign-key mappings
- Review date ranges and data types
- Check null, negative and zero quantities
- Review Data_Quality_Flag counts
- Reconcile SQL and Power BI totals

Interpretation limits

- Synthetic data, not official company records
- Root-cause labels do not prove causation
- Different fact-table grains can distort joins
- KPI totals depend on filters and definitions
- Dashboard findings need actual verification

Likely Questions & Suggested Answers

Is this official TVS data?

No. It is synthetic practice data used to demonstrate the analytics workflow.

Why use SQL and Power BI?

SQL stores, validates and aggregates data; Power BI helps model and communicate results.

Why separate fact tables?

Production and defect logs have different grains; careless joins can inflate totals.

How are KPIs validated?

Compare SQL outputs and Power BI measures using the same filters and definitions.

PROJECT REPOSITORY

Thank You

TVS Motor Manufacturing Quality, Defect & Production Analytics

GitHub repository

https://github.com/prithivikanth/prithivikanth-TVS-Motor-Manufacturing-Quality-Analytics

Prithivi Kanth S

Synthetic portfolio dataset • Verify all findings before presenting